



Modelling Financial Markets

Chapter 18: Systemic Risk and Networks



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-FI: BET Financial Investment companies index
- ▣ BET-NG: Bucharest Exchange Trading – Energy and Related Utilities (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNP: BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAC: Cotation Assistée en Continu (CAC 40 index) (the CAC 40 index of the Paris Stock Exchange)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DAX: Deutscher Aktienindex (the German stock index (Frankfurt Stock Exchange))
- ▣ DCC: Dynamic Conditional Correlation
- ▣ ECB: European Central Bank
- ▣ ES: Expected Shortfall
- ▣ EU: European Union
- ▣ EURIBOR: Euro Interbank Offered Rate
- ▣ FDIC: Federal Deposit Insurance Corporation
- ▣ FDR: False Discovery Rate
- ▣ FRM: Financial Risk Meter
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GAAP: Generally Accepted Accounting Principles (US)
- ▣ GACV: Generalized Approximate Cross-Validation
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HSBC: HSBC Holdings plc (originally the Hongkong and Shanghai Banking Corporation)

Acronyms used in this material (2/2)

- ▣ IFRS: International Financial Reporting Standards
- ▣ ING: ING Groep (Internationale Nederlanden Groep) (the Dutch banking group ING)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ LRMES: Long-Run Marginal Expected Shortfall
- ▣ MES: Marginal Expected Shortfall
- ▣ OLS: Ordinary Least Squares
- ▣ QML: Quasi-Maximum Likelihood
- ▣ ROBOR: Romanian Interbank Offer Rate
- ▣ ROTX: Romanian Traded Index
- ▣ SG: Société Générale (the French banking group Société Générale)
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ SVB: Silicon Valley Bank
- ▣ TENET: Tail-Event driven NETWORK risk
- ▣ US: United States
- ▣ VAR: Vector AutoRegression
- ▣ VIX: CBOE Volatility Index

Outline

Motivation

Exposure: MES and SRISK

Contribution: CoVaR

Connectedness: Spillover Networks

Tail Networks: the Financial Risk Meter

Climate Risk

Stress Testing

Case Study from Our Research: FRM@RO

AI for Scientific Discovery

Summary

References

What Is Systemic Risk?

From one bank to the system

- ▣ **Systemic risk**: the risk that the distress of one or several institutions impairs the financial system as a whole, and through it the real economy
- ▣ It is an **externality**: a bank does not pay for the losses its failure imposes on others
- ▣ Chapter 7 measured the risk **of** a portfolio; this chapter measures the risk a bank **adds to** or **takes from** the system

What we do in this chapter

- ▣ Measure the exposure of banks to market crashes: MES and SRISK
- ▣ Measure the contribution of a bank to system losses: CoVaR and ΔCoVaR
- ▣ Map the network of volatility spillovers (Diebold–Yilmaz) and tail links (the Financial Risk Meter)
- ▣ Design stress tests: historical, hypothetical and reverse; climate risk

Bank Runs: Northern Rock 2007 and Silicon Valley Bank 2023



Depositors queue outside a Northern Rock branch,
Brighton, 14 September 2007

Photo: [Dominic Alves \(2007\)](#); CC BY 2.0; [Wikimedia Commons](#)

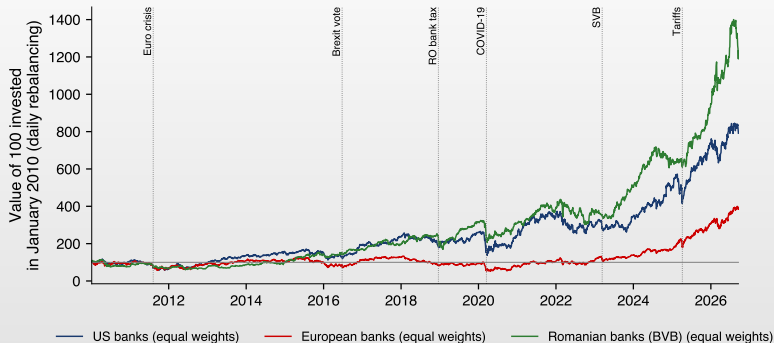
- ❑ Northern Rock (2007): a British mortgage lender hit by the freeze of funding markets after the US subprime losses; depositors queued to withdraw their savings
- ❑ SVB (2023): a digital run; the FDIC took the bank over on 10 March [FDIC \(2023\)](#); two days later all depositors of SVB and Signature Bank were protected on systemic risk grounds [Federal Reserve Board \(2023\)](#)
 - ▶ Section 7 measures the loss of a bank portfolio in that window
- ❑ Channels of contagion
 - ▶ **Funding runs:** depositors and lenders withdraw from any bank that looks similar
 - ▶ **Common exposures and fire sales:** forced selling lowers prices for every holder
 - ▶ **Counterparty links:** one bank's liability is another bank's asset, so a default passes losses to its creditors

Data Used in This Chapter

Group	Series	Market index
US banks	JPMorgan, Bank of America, Citigroup, Goldman Sachs, Morgan Stanley, Wells Fargo	S&P 500
European banks	Deutsche Bank, BNP Paribas, Santander, ING, HSBC	Euro Stoxx 50
Romanian banks	Banca Transilvania, BRD-Groupe Société Générale	BET
Volatility	the VIX index (implied volatility of the S&P 500)	

- Daily prices from January 2010 to 18 September 2026, adjusted for dividends and splits; 3 960 common trading days
- Prices are joined on common days first, then daily log returns $r_t = 100(\ln P_t - \ln P_{t-1})$, in %
- Romanian banks: days without trades removed; historical stress scenarios for the international banks start in 2007

Three Banking Systems, 2010–2026



- Equal-weighted portfolios of the banks of each region, rebalanced daily; vertical lines mark the episodes used in this chapter

Reading the Three Banking Systems

- ▣ 100 invested in January 2010 became 791 in US banks, 384 in European banks and 1233 in Romanian banks
 - ▶ Largest peak-to-trough fall: US 51% (2011-11-23), Europe 61% (2020-04-21), Romania 47% (2012-06-06)
- ▣ Annualised volatility of the portfolios: 29%, 30% and 23%, against 18% (S&P 500), 21% (Euro Stoxx 50) and 18% (BET)
- ▣ The falls are **simultaneous**: the euro crisis, COVID-19 and the 2023 bank failures hit all regions in the same weeks
- ▣ Romanian banks also have their own shock: the bank-tax announcement of December 2018

Conventions and a Map of Systemic Risk Measures

- Return X , loss $L = -X$; the level α is a **tail probability**: VaR 1%, CoVaR 1%, MES 5% (Chapter 7)
 - ▶ $\text{VaR}_\alpha(X) = -q_\alpha(X)$, with q_α the α -quantile; VaR, CoVaR and MES are positive numbers when they are losses
- **System** return X^{sys} : the equal-weighted portfolio of the other banks of the same region
- Three families of measures Billio, Getmansky, Lo & Pelizzon (2012)
 - ▶ **Exposure**: how much does the bank lose when the market or the system crashes? MES, SRISK, exposure CoVaR
 - ▶ **Contribution**: how much worse is the system when the bank is in distress? ΔCoVaR
 - ▶ **Connectedness**: through which links do shocks travel? Spillover networks, tail networks
- Survey of which measure rests on which theory: Benoit, Colliard, Hurlin & Pérignon (2017); textbooks: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017)

Recap: Motivation

- ▣ Systemic risk is about the system, not the single bank: it lives in the joint tail of many institutions
- ▣ Runs, common exposures, fire sales and counterparty links turn one failure into many
- ▣ Bank portfolios fall together in crises and are more volatile than the market
- ▣ We need measures of exposure, contribution and connectedness, each with its uncertainty

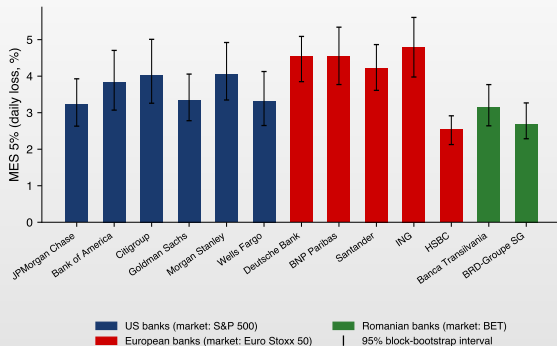
Marginal Expected Shortfall

- **MES** (Marginal Expected Shortfall) of bank i at level α [Acharya, Pedersen, Philippon & Richardson \(2017\)](#)
 - ▶ $\text{MES}_{i,\alpha} = -\mathbb{E}[X_i \mid X_m \leq q_\alpha(X_m)]$
 - ▶ The average loss of the bank on the worst α share of days for the market m
- Why “marginal”: if the conditioning variable is the portfolio itself, $m = \sum_i w_i X_i$, then $\text{ES}_\alpha(X_m) = \sum_i w_i \text{MES}_{i,\alpha}$ (Euler allocation, next slide)
 - ▶ Our markets (S&P 500, Euro Stoxx 50, BET) are **not** portfolios of these 13 banks: here MES measures **exposure**, not a share of the index ES
- Historical estimate: average $-X_i$ over the days with $X_m \leq \hat{q}_\alpha(X_m)$; with $\alpha = 5\%$ and 3 960 days, about 200 days
- Market m : S&P 500 for US banks, Euro Stoxx 50 for European banks, BET for Romanian banks

MES as an Euler Contribution: Proof

- Portfolio $X_p = \sum_i w_i X_i$ with a continuous distribution; $ES_\alpha(X_p) = -\mathbb{E}[X_p \mid X_p \leq q_\alpha(X_p)]$
 - ▶ ES_α is positively homogeneous of degree 1 in w , so by Euler's theorem $ES_\alpha(X_p) = \sum_i w_i \partial ES_\alpha / \partial w_i$
 - ▶ Differentiating $-\frac{1}{\alpha} \mathbb{E}[X_p \mathbf{1}\{X_p \leq q_\alpha(w)\}]$: the term in $\partial q_\alpha / \partial w_i$ is $-\frac{1}{\alpha} q_\alpha f_p(q_\alpha) \partial q_\alpha / \partial w_i + \frac{1}{\alpha} q_\alpha f_p(q_\alpha) \partial q_\alpha / \partial w_i = 0$
 - ▶ Hence $\partial ES_\alpha / \partial w_i = -\mathbb{E}[X_i \mid X_p \leq q_\alpha(X_p)] = MES_{i,\alpha}$ with $m = p$ [Gouriéroux, Laurent & Scaillet \(2000\)](#), [Scaillet \(2004\)](#)
- Check on our data: equal-weighted portfolio of the six US banks, $\alpha = 5\%$ (198 tail days)
 - ▶ $ES_{5\%}$ of the portfolio = 4.32% = average of the six MES conditional on the portfolio (4.32%): exact, also for the historical estimator
 - ▶ Conditioning on the S&P 500 instead gives an average MES of 3.63% (JPMorgan: 3.22% vs 3.85%): the identity fails when $m \neq p$
- Under the bivariate Normal distribution with zero means, $MES_{i,\alpha} = \beta_i ES_\alpha(X_m)$: a beta times a constant (Seminar 18, A3)

MES 5% of Thirteen Banks



- Average daily loss of each bank on the worst 5% of days for its market index, 2010–2026; 95% intervals from a moving-block bootstrap (blocks of 20 days)

Reading the MES

- ▣ US banks: from 3.22% (JPMorgan) to 4.04% (Morgan Stanley); European banks up to 4.81% (ING)
 - ▶ On the same days the market lost on average 2.72% (S&P 500) and 3.16% (Euro Stoxx 50): banks lose more than the market
- ▣ HSBC is the exception in Europe (2.54%, below the index): a London listing with Asian business
- ▣ Banca Transilvania 3.14% and BRD 2.69% on BET crash days; BET itself lost 2.70%
- ▣ Overlapping marginal intervals are not a test of equality: bank-to-bank differences need a paired test on joint bootstrap draws (Section 3)

From MES to a Crisis: LRMES

- **LRMES (Long-Run MES):** the expected fall of the bank's equity in a crisis, a market fall below C over h days
 - ▶ $\text{LRMES}_{i,t} = -\mathbb{E}_t [R_{i,t+1:t+h} \mid R_{m,t+1:t+h} < C]$, arithmetic h -day returns
 - ▶ Acharya, Engle & Richardson (2012) (Section I): $C = -40\%$ over six months; Brownlees & Engle (2017) (Section 1.1): $C = -10\%$ over $h = 22$ days
- **Shortcut Acharya, Engle & Richardson (2012), Section I,** for “versions of the model where the simulation is not yet implemented”:

$$\text{LRMES}_i \approx 1 - \exp(-18 \times \text{MES}_i^{2\%})$$
 - ▶ $\text{MES}_i^{2\%}$: average daily loss of the bank on days when the market falls by more than 2% (in decimals); example $\text{MES}^{2\%} = 3\%$ gives $1 - e^{-0.54} = 41.7\%$
 - ▶ The 18 is a calibration constant; Seminar 18, A4 derives where it comes from
- **Shortcut on our banks:** from 37% (HSBC) to 58% (ING); JPMorgan 48%, Banca Transilvania 52%
 - ▶ The market fell by more than 2% on 138 days in the US, 196 in the euro area and 111 on the BVB

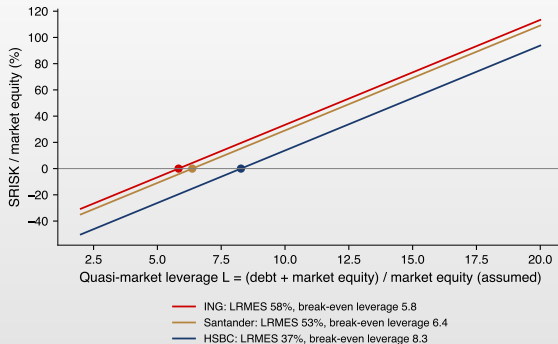
SRISK: the Capital Shortfall in a Crisis

- A bank must keep equity of at least a share k of its quasi-assets (book debt D plus market equity W) [Brownlees & Engle \(2017\)](#), Section 1.1, Eq. (1)
 - ▶ $SRISK_i = k D_i - (1 - k)(1 - LRMES_i) W_i$, assuming debt is not renegotiated in the crisis
 - ▶ $k = 8\%$ for US firms [Brownlees & Engle \(2017\)](#); $k = 5.5\%$ for European firms, whose IFRS balance sheets report larger assets than US GAAP [Engle, Jondeau & Rockinger \(2015\)](#)
- Positive SRISK = shortfall; the SRISK of the system is the sum of the positive values; [V-Lab, New York University Stern School of Business](#) publishes it for large banks with their balance sheets
- Example: $D = 900$, $W = 100$, $LRMES = 41.7\%$
 - ▶ $SRISK = 0.08 \times 900 - 0.92 \times (1 - 41.7\%) \times 100 = 72 - 53.6 = 18.4$
- SRISK combines a market measure (LRMES) with a balance-sheet measure (leverage)

SRISK per Unit of Equity and the Break-Even Leverage

- Divide by W and write the quasi-market leverage $L = (D + W)/W$
 - ▶ $\frac{\text{SRISK}}{W} = k(L - 1) - (1 - k)(1 - \text{LRMES})$
- SRISK is positive once leverage exceeds the **break-even leverage**
 - ▶ $L^* = 1 + \frac{(1 - k)(1 - \text{LRMES})}{k}$
 - ▶ Example above: $L^* = 7.7$, while the bank has $L = 10$: a shortfall
- Our data contain prices, not balance sheets: we report L^* from the estimated LRMES and treat leverage as an **assumption**
- A higher LRMES lowers L^* : the bank can afford less debt before a crisis creates a shortfall

SRISK as a Function of Leverage



- Banks with the highest, median and lowest LRMES; leverage on the horizontal axis is assumed, not observed; dots mark L^*

Reading the Break-Even Leverage

- ▣ With $k = 8\%$: $L^* = 5.8$ for ING, 6.4 for Santander, 8.3 for HSBC
 - ▶ US: JPMorgan 7.0, Morgan Stanley 6.0; Romania: Banca Transilvania 6.5, BRD 7.1
- ▣ A bank whose debt is more than about 5–7 times its market equity would need new capital in a 40% market crash
- ▣ With $k = 5.5\%$ the thresholds rise (ING 8.2, HSBC 11.9): the prudential ratio is a policy choice, and SRISK depends on it
- ▣ Limits: LRMES is an extrapolation from daily data, and market equity itself falls in a crisis

LRMES as Published: GJR–GARCH, DCC and Simulation

- Model of [Brownlees & Engle \(2017\)](#), Section 1.2: zero-mean log returns of bank and market, GJR–GARCH(1,1) volatilities [Glosten, Jagannathan & Runkle \(1993\)](#) and a DCC(1,1) correlation [Engle \(2002\)](#), two-step QML
 - ▶ $\sigma_t^2 = \omega + (\alpha + \gamma \mathbf{1}\{r_{t-1} < 0\}) r_{t-1}^2 + \beta \sigma_{t-1}^2$, for the bank and for the market
 - ▶ $Q_t = (1 - a - b) S + a \epsilon_{t-1} \epsilon_{t-1}^\top + b Q_{t-1}$, $\rho_t = Q_{12,t} / \sqrt{Q_{11,t} Q_{22,t}}$
- Simulation (Appendix A): resample pairs of standardised innovations $(\xi_{it}, \epsilon_{mt})$ (next slide), filter h days ahead from today's σ^2 and ρ , keep the paths with $R_{m,t+1:t+h} < C$
 - ▶ LRMES = minus the average bank return over the crisis paths; resampling pairs keeps the tail dependence that the Normal distribution misses
- Our run: data to 2026-09-18; 200 000 paths of 22 days per market ($C = -10\%$), 400 000 paths of 126 days ($C = -40\%$)
 - ▶ Probability of the 22-day crisis today: 1.3% (S&P 500), 4.3% (Euro Stoxx 50), 4.0% (BET); on 31 March 2020: 12%, 19%, 8%

LRMES Simulation: Innovations and Recursions

- Standardised residuals $\epsilon_{it} = r_{it}/\sigma_{it}$, $\epsilon_{mt} = r_{mt}/\sigma_{mt}$; S is their sample correlation matrix, the long-run target of Q_t
 - ▶ Bank innovation orthogonal to the market: $\xi_{it} = (\epsilon_{it} - \rho_t \epsilon_{mt})/\sqrt{1 - \rho_t^2}$ [Brownlees & Engle \(2017\)](#), Appendix A
- Each simulated day s : draw a past date τ , take the pair $(\xi_{i\tau}, \epsilon_{m\tau})$ and rebuild the bank shock with the **simulated** correlation
 - ▶ $\epsilon_{i,t+s} = \rho_{t+s} \epsilon_{m\tau} + \sqrt{1 - \rho_{t+s}^2} \xi_{i\tau}$, then update σ^2 and Q with the new returns
- Restrictions: $\omega > 0$, $\alpha \geq 0$, $\alpha + \gamma \geq 0$, $\beta \geq 0$, $\alpha + \gamma/2 + \beta < 1$; $a, b > 0$, $a + b < 1$, so that $\sigma_t^2 > 0$ and Q_t is positive definite
 - ▶ Initialisation: $Q_1 = S$ in estimation; the simulation starts from σ_{T+1}^2 and Q_{T+1} filtered to the last date

LRMES by Simulation: Thirteen Banks

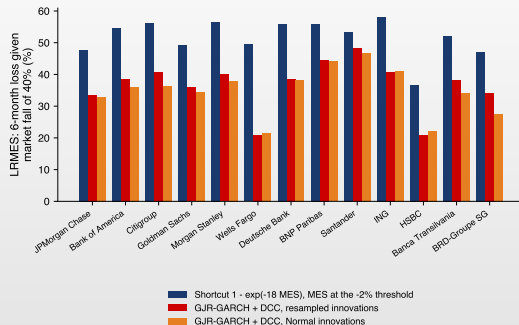
Bank	LRMES today	90% range	LRMES 31.03.2020	ρ_T	L^*
JPMorgan	10.6	[-2, 25]	23.7	0.53	11.3
Bank of America	14.3	[-4, 35]	23.2	0.44	10.9
Citigroup	13.0	[-1, 28]	32.8	0.54	11.0
Goldman Sachs	16.5	[1, 34]	25.0	0.65	10.6
Morgan Stanley	16.3	[1, 32]	27.2	0.65	10.6
Wells Fargo	6.7	[-8, 24]	22.9	0.32	11.7
Deutsche Bank	14.4	[2, 28]	25.7	0.66	10.8
BNP Paribas	15.6	[2, 33]	28.5	0.68	10.7
Santander	16.6	[3, 34]	26.4	0.76	10.6
ING	12.5	[1, 27]	32.9	0.68	11.1
HSBC	8.3	[-2, 19]	10.9	0.58	11.5
Banca Transilvania	12.4	[-2, 34]	17.2	0.74	11.1
BRD-Groupe SG	11.4	[-2, 28]	15.8	0.64	11.2

- LRMES (%) for a market fall below -10% in 22 days [Brownlees & Engle \(2017\)](#); 90% range: 5% and 95% quantiles of the bank's loss over the crisis paths; ρ_T : last DCC correlation; L^* with $k = 8\%$

Reading LRMES by Simulation

- LRMES is **conditional**: it moves with today's volatility and correlation
 - ▶ JPMorgan: 10.6% today, 23.7% with the information of 31 March 2020; Deutsche Bank 14.4% and 25.7%
- The 90% range is wide: LRMES is an average over crisis paths, and the capital shortfall of a single crisis can be much larger [Brownlees & Engle \(2017\)](#), Eq. (2)
- The range reflects the crisis scenarios only: parameter uncertainty (GJR and DCC estimates) would widen it further
- With a one-month, -10% crisis, LRMES is about a third of the six-month, -40% numbers of [Acharya, Engle & Richardson \(2012\)](#): the scenario, not only the bank, sets the level

Model Risk: LRMES under Three Specifications



- Six-month crisis, market fall of 40% Acharya, Engle & Richardson (2012); shortcut $1 - e^{-18\text{MES}^{2\%}}$ vs simulation from GJR-GARCH + DCC with resampled or Normal innovations, data to 2026-09-18

Reading the Model Risk of LRMES

- Simulated LRMES divided by the shortcut: between 0.42 and 0.90 across the 13 banks; rank correlation 0.77
 - ▶ JPMorgan: 48% (shortcut), 33% (resampled), 33% (Normal innovations); ING: 58%, 41%, 41%
- The shortcut uses the whole 2010–2026 history; the simulation starts from today's volatility and correlation
 - ▶ A 40% fall in six months is rare today: probability 0.29% (S&P 500), 0.96% (Euro Stoxx 50), 0.30% (BET) under the model
- Resampled vs Normal innovations: rank correlation 0.95; the innovation distribution changes levels more than rankings
- Same banks, same data, different defensible models, different numbers: the “model risk of risk models” [Danielsson, James, Valenzuela & Zer \(2016\)](#)

Recap: MES and SRISK

- MES is the bank's average loss on the market's worst days; weighted by portfolio shares, the MES add up to the ES of a portfolio only when they condition on that same portfolio
- LRMES extrapolates MES to a six-month crisis; SRISK turns it into the capital shortfall
- European banks have the largest MES in our sample; bank and region rankings carry wide intervals
- As published, LRMES is simulated from GJR-GARCH + DCC; the $1 - e^{-18\text{MES}}$ shortcut differs by bank, and the scenario (h , C) sets the level
- SRISK needs balance-sheet data; without it we can report only the break-even leverage

CoVaR: the VaR of the System Given a Bank in Distress

- **CoVaR** of the system conditional on bank i at its VaR [Adrian & Brunnermeier \(2016\)](#)
 - ▶ $P\left(X^{sys} \leq -\text{CoVaR}_{\alpha}^{sys|i} \mid X_i = -\text{VaR}_{i,\alpha}\right) = \alpha$
 - ▶ The loss of the system that is exceeded with probability α on a day when bank i is at its own VaR
- **ΔCoVaR** : the change from the median state of the bank to its distress state
 - ▶ $\Delta\text{CoVaR}_{\alpha}^{sys|i} = \text{CoVaR}_{\alpha}^{sys|X_i=q_{\alpha}} - \text{CoVaR}_{\alpha}^{sys|X_i=q_{50\%}}$
- **ΔCoVaR** measures the **contribution** of bank i to the tail risk of the system
- Reversing the roles gives the **exposure CoVaR**: the VaR of bank i when the system is in distress

Quantile Regression

- The α -quantile of Y given x is linear: $q_\alpha(Y | x) = a_\alpha + b_\alpha x$ [Koenker & Bassett \(1978\)](#)
 - ▶ $(\hat{a}_\alpha, \hat{b}_\alpha) = \arg \min_{a,b} \sum_t \rho_\alpha(y_t - a - b x_t)$, $\rho_\alpha(u) = u(\alpha - 1\{u < 0\})$
 - ▶ ρ_α is the **check function**: it weights negative residuals by $1 - \alpha$ and positive ones by α ; $\alpha = 50\%$ gives the median regression
- Asymptotics for a fixed α , with $x_t = (1, X_{i,t})^\top$ and f_t the conditional density of Y at its α -quantile
 - ▶ $\sqrt{n}(\hat{\theta}_\alpha - \theta_\alpha) \Rightarrow N\left(0, \alpha(1 - \alpha) D_1^{-1} D_0 D_1^{-1}\right)$, $D_0 = \mathbb{E}[x_t x_t^\top]$, $D_1 = \mathbb{E}[f_t x_t x_t^\top]$
 - ▶ Assumes independent observations (or scores $x_t(\alpha - 1\{u_t < 0\})$ that are a martingale difference), $f_t > 0$ near the quantile and D_0, D_1 positive definite; with serially dependent scores D_0 needs a long-run (HAC) estimate or a block bootstrap
 - ▶ The **sparsity** $1/f_t$ must be estimated (kernel or bandwidth); at $\alpha = 1\%$ few observations lie near the quantile, so this estimate is noisy
- The slope b_α can change with α : the bank may matter more for the system in the tail than in the centre; least squares gives only the mean slope

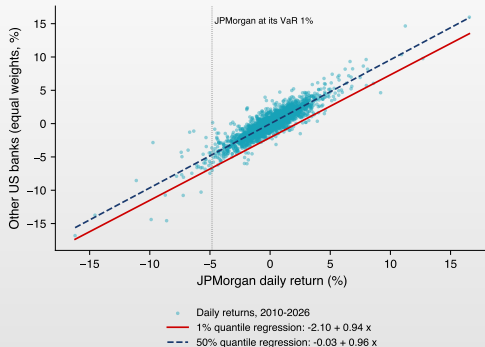
Estimating CoVaR in Three Steps

- ▣ 1. Quantile regression of the system on the bank at level α
 - ▶ $\hat{q}_\alpha(X^{sys} | X_i) = \hat{a}_\alpha + \hat{b}_\alpha X_i$
- ▣ 2. Plug in the bank's own quantiles: $\hat{q}_\alpha(X_i)$ (distress) and $\hat{q}_{50\%}(X_i)$ (median)
 - ▶ $\text{CoVaR}_\alpha^{sys|i} = - \left(\hat{a}_\alpha + \hat{b}_\alpha \hat{q}_\alpha(X_i) \right)$
- ▣ 3. Take the difference
 - ▶ $\Delta \text{CoVaR}_\alpha^{sys|i} = \hat{b}_\alpha (\hat{q}_{50\%}(X_i) - \hat{q}_\alpha(X_i))$
 - ▶ Only the slope matters: the intercept cancels
- ▣ Uncertainty: moving-block bootstrap of the days (blocks of 20 days keep volatility clusters together), 999 resamples

Distress at or below VaR: the Girardi–Ergün CoVaR

- Girardi & Ergün (2013) condition on the bank being at most at its VaR, not exactly at it
 - ▶ $P\left(X^{sys} \leq -\text{CoVaR}_{\alpha}^{\leq} \mid X_i \leq -\text{VaR}_{i,\alpha}\right) = \alpha \iff P\left(X^{sys} \leq -\text{CoVaR}_{\alpha}^{\leq}, X_i \leq -\text{VaR}_{i,\alpha}\right) = \alpha^2$
 - ▶ Benchmark state: a one-standard-deviation event $|X_i - \mu_i| \leq \sigma_i$ instead of the median
- Why it matters
 - ▶ It includes more severe distress than the VaR day itself, and it is monotone in the dependence parameter
 - ▶ It can be backtested: the joint hit $\{X^{sys} \leq -\text{CoVaR}^{\leq}, X_i \leq -\text{VaR}_i\}$ has probability α^2
- Historical estimate at $\alpha = 5\%$: the 5% quantile of the system on the 198 days with $X_i \leq \hat{q}_{5\%}(X_i)$
 - ▶ JPMorgan: $\text{CoVaR}_{5\%}^{\leq} = 7.70\%$ vs the quantile-regression $\text{CoVaR}_{5\%} = 3.81\%$; eleven banks: 7.3–8.2% vs 3.6–4.8%
 - ▶ About twice as large: averaging over the whole tail of the bank is a harsher conditioning event (Seminar 18, A2 derives this under the Normal distribution)

Quantile Regression: JPMorgan and the Other US Banks

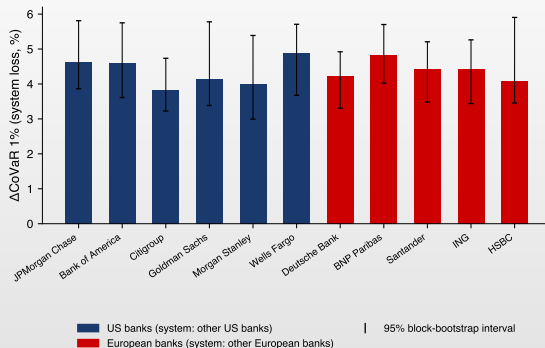


- Daily returns 2010–2026 with the 1% and 50% quantile regression lines; the dotted line marks JPMorgan at its VaR 1%

Reading the JPMorgan Example

- 1% line: $\hat{q}_{1\%} = -2.10 + 0.94 X_i$; median line: $-0.03 + 0.96 X_i$
 - ▶ Similar slopes: the 1% line is shifted down by the tail of the other banks, not steeper
- JPMorgan's VaR 1% is 4.83%; on such a day the other US banks have CoVaR 1% = 6.64%
 - ▶ Unconditional VaR 1% of the system: 5.07%; with JPMorgan at its median: 2.03%
- $\Delta\text{CoVaR}_{1\%} = 4.62\%$: JPMorgan in distress adds about 4.62 percentage points to the 1% loss of the other US banks
- This is association, not causation: a common shock hits JPMorgan and the others on the same day

ΔCoVaR 1% of Eleven Banks



□ System: equal-weighted portfolio of the other banks of the same region; 95% moving-block bootstrap intervals

Reading ΔCoVaR

- All eleven banks lie between 3.81% (Citigroup) and 4.89% (Wells Fargo); the 95% intervals are 1.5–2.4 points wide
 - ▶ E.g. Citigroup [3.23, 4.74], Wells Fargo [3.68, 5.71]: the marginal intervals overlap
- Overlap is **not** a test: two estimates computed on the same days are dependent (in our bootstrap draws strongly positively correlated), so the difference can be tighter than either interval
 - ▶ The next two slides test the differences on joint bootstrap draws
- At $\alpha = 5\%$ the values are smaller (JPMorgan 2.62%) and the ranking changes: Spearman correlation between the 1% and 5% rankings 0.10
 - ▶ A ranking that changes with α is not a robust basis for regulation

Testing ΔCoVaR : Three Hypotheses

- (1) No contribution, $H_0: b_\alpha = 0$: Wald test with the sandwich variance of the previous slides [Koenker & Bassett \(1978\)](#)
 - ▶ Tests of linear hypotheses across quantiles use the joint asymptotics of $(\hat{b}_\alpha, \hat{b}_{50\%})$ [Koenker & Bassett \(1982\)](#)
- (2) The tail link equals the central link, $H_0: b_\alpha = b_{50\%}$: is the bank special in crises?
 - ▶ We estimate the variance of $\hat{b}_\alpha - \hat{b}_{50\%}$ from 999 moving-block bootstrap draws that keep the pairs of days together
- (3) Equal contributions, $H_0: \Delta\text{CoVaR}_i = \Delta\text{CoVaR}_j$: the same resampled days for **all** banks give the joint distribution
 - ▶ $z_{ij} = (\widehat{\Delta\text{CoVaR}_i} - \widehat{\Delta\text{CoVaR}_j}) / \widehat{\text{se}}^*(\cdot)$ for each pair; Holm correction over the pairs [Holm \(1979\)](#); a joint Wald test that all are equal
 - ▶ Significance and dominance tests for systemic importance: [Castro & Ferrari \(2014\)](#)

Testing ΔCoVaR 1%: Results

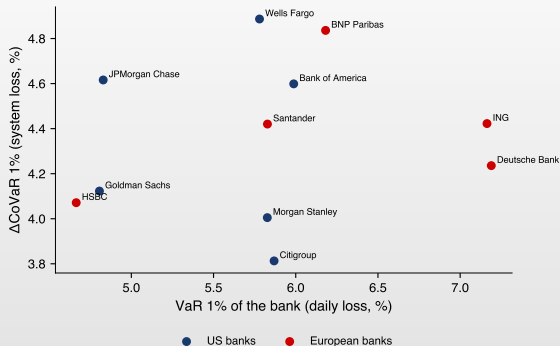
Bank	ΔCoVaR	$\hat{b}_{1\%}$	z	$\hat{b}_{50\%}$	$\hat{b}_{1\%} - \hat{b}_{50\%}$	p
JPMorgan	4.62	0.94	6.9	0.96	-0.02	0.66
Bank of America	4.60	0.76	6.5	0.78	-0.02	0.73
Citigroup	3.81	0.64	4.7	0.76	-0.11	0.00
Goldman Sachs	4.12	0.85	5.6	0.84	0.00	0.99
Morgan Stanley	4.00	0.68	4.5	0.73	-0.05	0.28
Wells Fargo	4.89	0.84	4.3	0.78	0.06	0.17
Deutsche Bank	4.24	0.59	4.1	0.60	-0.02	0.67
BNP Paribas	4.84	0.77	6.5	0.74	0.03	0.37
Santander	4.42	0.75	4.5	0.69	0.07	0.07
ING	4.42	0.61	4.4	0.71	-0.10	0.00
HSBC	4.07	0.87	4.0	0.86	0.01	0.96

- z : kernel (sandwich) t -ratio of $\hat{b}_{1\%}$, valid only for serially independent scores: every bank contributes; p : bootstrap test of $b_{1\%} = b_{50\%}$, significant for 2 of 11 banks (2 after Holm)
- ▶ Equal contributions within a region: 4 of 25 pairs differ at 5%, 0 after Holm; joint Wald $p = 0.01$ (US), 0.31 (Europe)
 - ▶ At 5%: 3 of 25 pairs differ, 0 after Holm; joint Wald $p = 0.014$ (US), 0.046 (Europe)

Extremal Quantiles: Why $\alpha = 1\%$ Is Hard

- The effective tail sample is αn : 40 days at 1% and 198 at 5% out of 3 960
 - ▶ The Normal approximation above assumes $\alpha n \rightarrow \infty$ fast; with αn of a few dozen it can be poor, and extremal-quantile theory (extreme-value limits, subsampling) is the alternative built for this case [Chernozhukov & Fernández-Val \(2011\)](#)
- In our data the block-bootstrap s.e. of $\Delta\text{CoVaR } 1\%$ is about 0.6 times the kernel-based s.e. (median over banks): the two routes disagree by a large factor
 - ▶ The kernel s.e. rests on a sparsity estimate from the few observations near the 1% quantile; the bootstrap redraws the same few tail days: neither is reliable when αn is this small
- Practical rule: report $\alpha = 5\%$ and $\alpha = 1\%$ with αn ; expect the 1% ranking to be the noisier one
 - ▶ The low Spearman correlation between the 1% and 5% rankings (0.10) may reflect estimation noise, genuine differences between the two tail levels, or both

VaR and ΔCoVaR Measure Different Things



□ Horizontal: VaR 1% of the bank; vertical: its ΔCoVaR 1%, eleven banks, 2010–2026

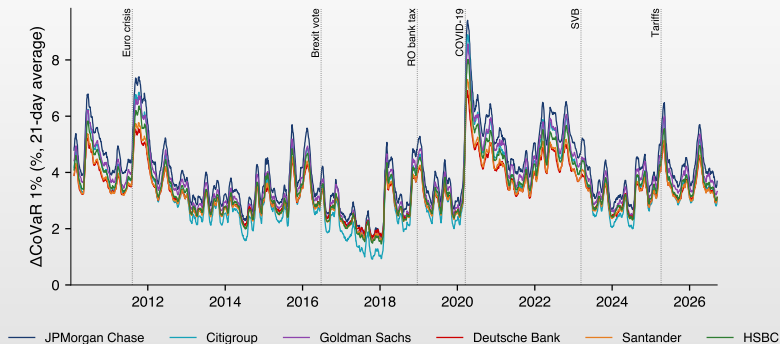
Reading VaR versus ΔCoVaR

- Spearman rank correlation between VaR and ΔCoVaR : 0.16
 - ▶ Deutsche Bank and ING have the largest VaR (7.19%, 7.16%) but only average ΔCoVaR
 - ▶ JPMorgan has one of the smallest VaR values (4.83%) and one of the largest ΔCoVaR values
- This is the central message of [Adrian & Brunnermeier \(2016\)](#): regulating each bank by its own VaR ignores its contribution to the system
- Caveat: with eleven banks and overlapping intervals, the correlation itself is very uncertain (Seminar 18, B2)

Time-Varying ΔCoVaR with State Variables

- State variables M_{t-1} : log VIX, the return of the regional index and of the system, all lagged one day [Adrian & Brunnermeier \(2016\)](#)
 - ▶ $q_\alpha(X_{i,t} \mid M_{t-1}) = a_i + c_i^\top M_{t-1}$ at α and at 50%
 - ▶ $q_\alpha(X_t^{\text{sys}} \mid X_{i,t}, M_{t-1}) = a + b X_{i,t} + c^\top M_{t-1}$
- Then $\Delta\text{CoVaR}_{\alpha,t} = \hat{b} (\hat{q}_{50\%,t}(X_i) - \hat{q}_{\alpha,t}(X_i))$
 - ▶ It moves with the bank's conditional VaR, which moves with the VIX
- The slope $\hat{b}$ is constant: time variation comes from the state, not from the link

$\Delta\text{CoVaR 1\%}$ over Time

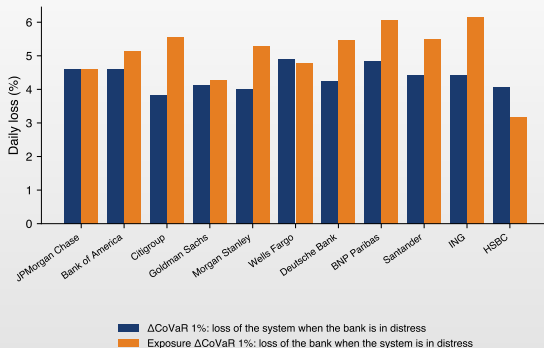


- Six banks, 21-day moving averages of the daily $\Delta\text{CoVaR 1\%}$ estimated with lagged state variables

Reading ΔCoVaR over Time

- ▣ JPMorgan: 2.16% on average in 2017, 9.06% in March 2020; peak 10.99% on 2020-03-18
 - ▶ Deutsche Bank: 2.04% in 2017 and 6.65% in March 2020
- ▣ The six lines move together: the common state variables drive them
- ▣ Market-based ΔCoVaR is low in calm periods, exactly when risk may be building up in balance sheets, which market data do not see
- ▣ [Adrian & Brunnermeier \(2016\)](#) therefore propose a forward ΔCoVaR , predicted from leverage, size and maturity mismatch

Contribution versus Exposure



- Blue: $\Delta\text{CoVaR 1\%}$ (loss of the system when the bank is in distress); orange: exposure $\Delta\text{CoVaR 1\%}$ (loss of the bank when the system is in distress)

Reading Contribution versus Exposure

- The direction matters: the two numbers differ for most banks
 - ▶ ING: contribution 4.42%, exposure 6.14%; BNP Paribas: 4.84% and 6.07%
 - ▶ HSBC: contribution 4.07%, exposure only 3.17%
- Exposure is larger than contribution for most European banks: the rise in the bank's tail loss when the system is in distress exceeds the rise in the system's tail loss when the bank is in distress; both are associations, not causal effects
- Exposure matters for the bank's own capital; contribution matters for the regulator of the system

Backtesting CoVaR and MES

- Systemic measures are conditional risk measures, so they need **conditional** backtests Banulescu-Radu, Hurlin, Leymarie & Scaillet (2021)
 - ▶ For $\text{CoVaR}^{\leq}$: hit $I_t = 1\{X_t^{\text{sys}} \leq -\text{CoVaR}_t^{\leq}, X_{i,t} \leq -\text{VaR}_{i,t}\}$ with $P(I_t = 1) = \alpha^2$ under H_0 if both the bank VaR and the conditional quantile are calibrated; Kupiec and Christoffersen tests on I_t (Chapter 8), plus a separate check of the bank VaR rate
 - ▶ For MES: the cumulative joint violation process of Banulescu-Radu, Hurlin, Leymarie & Scaillet (2021)
- Our out-of-sample check: historical $\text{CoVaR}_{5\%}^{\leq}$ on an expanding window from day 1 000, 2 960 forecast days per bank
 - ▶ Expected joint hits 7.4; observed 5–9 across the eleven banks (JPMorgan 9); smallest Kupiec $p = 0.35$: no rejection
 - ▶ Power is limited: if the true joint rate were twice α^2 , the 5% Kupiec test would reject in 62% of samples of this length with independent hits, so even a doubling is missed about one time in three

How Much Can We Trust a Ranking?

- Rankings of banks by different systemic measures disagree, and each ranking has wide sampling uncertainty [Nucera, Schwaab, Koopman & Lucas \(2016\)](#)
 - ▶ They propose combining measures and reporting rank uncertainty instead of a single ordering
- Model risk: the same bank's risk measure varies widely across estimation methods [Danielsson, James, Valenzuela & Zer \(2016\)](#)
 - ▶ In this chapter: LRMES shortcut vs simulation (ratio 0.42–0.90), quantile-regression vs Girardi–Ergün CoVaR (about $\times 2$), $\alpha = 1\%$ vs 5% (Spearman 0.10)
- What survives: large, tested differences (regions; the tail slope of some banks), not the order of eleven close point estimates

Recap: CoVaR

- CoVaR is the VaR of the system given a bank in distress; ΔCoVaR is the bank's contribution
- Estimated by quantile regression; only the tail slope $\hat{b}_\alpha$ and the bank's quantiles matter
- VaR and ΔCoVaR rank banks differently; differences between banks must be tested on joint draws, with a multiplicity correction
- With state variables, ΔCoVaR follows market volatility: it rises in crises and says little in calm times

From Pairs to Networks

- ▣ CoVaR looks at one bank and one system; a **network** looks at all pairs at once
 - ▶ Nodes: banks; directed edges: how much of the future uncertainty of bank i comes from shocks to bank j
- ▣ Granger-causality networks of banks, insurers, hedge funds and brokers Billio, Getmansky, Lo & Pelizzon (2012)
- ▣ Variance-decomposition networks Diebold & Yilmaz (2009), Diebold & Yilmaz (2012), Diebold & Yilmaz (2014): the approach of this section
- ▣ Tail networks from quantile regressions: TENET Härdle, Wang & Yu (2016)

The Forecast Error Variance Decomposition

- Vector autoregression (VAR) of order p for N series y_t : $y_t = c + \sum_{l=1}^p A_l y_{t-l} + \varepsilon_t$, $\text{Cov}(\varepsilon_t) = \Sigma$
 - ▶ Moving-average form: $y_t = \mu + \sum_{h \geq 0} \Phi_h \varepsilon_{t-h}$, $\Phi_0 = I$, $\Phi_h = \sum_{l=1}^p A_l \Phi_{h-l}$
- The H -day forecast error of y_i is a sum of shocks to all series: its variance can be split by source
- Orthogonal (Cholesky) shocks depend on the ordering of the series; the **generalized** decomposition does not [Koop, Pesaran & Potter \(1996\)](#), [Pesaran & Shin \(1998\)](#)
 - ▶ $\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i^\top \Phi_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i^\top \Phi_h \Sigma \Phi_h^\top e_i}$, normalised: $\tilde{\theta}_{ij} = \theta_{ij} / \sum_j \theta_{ij}$

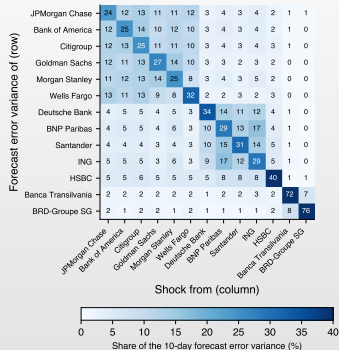
The Connectedness Table

- $\tilde{\theta}_{ij}$ (in %): the share of the H -day forecast error variance of i due to shocks in j ; each row sums to 100 [Diebold & Yilmaz \(2014\)](#)
 - ▶ **From others:** $F_i = \sum_{j \neq i} \tilde{\theta}_{ij}$; **to others:** $T_j = \sum_{i \neq j} \tilde{\theta}_{ij}$
 - ▶ **Net:** $T_i - F_i$ (positive = net transmitter); **pairwise net:** $\tilde{\theta}_{ji} - \tilde{\theta}_{ij}$
- **Total connectedness:** $C = \frac{1}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$, the average share of variance that comes from others
 - ▶ Rows of θ_{ij} do not sum to one before normalisation because generalized shocks are correlated (Seminar 18, A5 derives this)
 - ▶ A pairwise net link $\tilde{\theta}_{ji} - \tilde{\theta}_{ij} > 0$ means that i transmits net to j and draws an arrow $i \rightarrow j$: the network of the next slides

What We Connect: Daily Range Volatility

- **Parkinson volatility** from the daily high H_t and low L_t [Parkinson \(1980\)](#)
 - ▶ $\hat{\sigma}_t^2 = \frac{(\ln H_t - \ln L_t)^2}{4 \ln 2}$, annualised $100\sqrt{252 \hat{\sigma}_t^2}$, in logs
 - ▶ Unbiased for the variance of a Brownian motion without drift; more efficient than the squared return
- Volatility is persistent and therefore forecastable, while daily returns are close to unpredictable: a natural object for a VAR
- Our model: VAR on the log volatility of 13 banks, order chosen by the Bayesian information criterion ($p = 2$), horizon $H = 10$ days
 - ▶ Time zones: US banks close after European ones; the VAR lags capture the next-day transmission

The Connectedness Table of Thirteen Banks

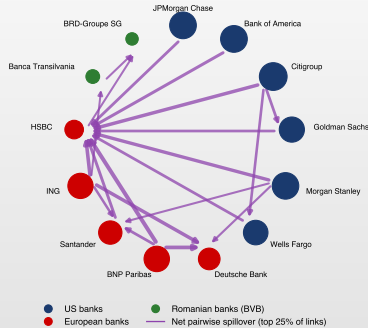


□ Row i , column j : share (%) of the 10-day forecast error variance of bank i due to shocks in bank j , full sample

Reading the Connectedness Table

- Total connectedness $C = 63.9\%$: on average two thirds of a bank's volatility risk comes from the other banks
 - ▶ On returns instead of volatility: $C = 75.4\%$
- Two regional blocks: a US bank receives 84.0% from US shocks and a European bank 72.7% from European shocks, own shock included
 - ▶ Across the Atlantic: a European bank receives 26.2% from US banks, a US bank 15.0% from European ones
- Romanian banks are largely their own: own share 72.3% (Banca Transilvania) and 76.1% (BRD); they receive 10.3% from US and 8.0% from European banks

The Net Spillover Network



- Arrow from j to i when j transmits net to i (largest 25% of pairwise net spillovers); node size grows with the spillover to others

Reading the Network

- Net transmitters: Citigroup (+19.0), Bank of America (+10.9), Morgan Stanley (+9.9), JPMorgan (+9.8)
 - ▶ To others: Citigroup 93.8%, JPMorgan 86.2% (the sum of a column can exceed 100)
- Net receivers: HSBC (−23.6), BRD (−13.0), Deutsche Bank (−12.6), Banca Transilvania (−12.2)
 - ▶ HSBC receives from almost every bank: it trades in London, between Asia and New York
- Within Europe, BNP Paribas and ING transmit to Deutsche Bank and Santander
- A full-sample network averages quiet and crisis years: the next chart lets it move

Connectedness over Time



- Top: total connectedness on moving 250-day windows ($\text{VAR}(2)$, $H = 10$, every 5 days); bottom: net spillover summed by region

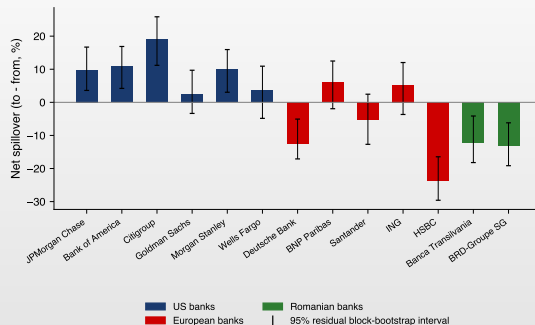
Reading Connectedness over Time

- ▣ The total index ranges from 47.4% (2024-07-24) to 79.3% (2020-08-27); mean 62.0%
 - ▶ COVID-19: from 61.8% in 2019 to 78.4% in spring 2020; it jumps within days and decays over a year
 - ▶ After the 2023 bank failures 66.0%; the calm year 2024 53.5%
- ▣ US banks are net transmitters in 89% of the windows (average net +48.4)
 - ▶ European banks were net transmitters during the euro crisis of 2011–2012 and mostly net receivers afterwards
- ▣ Romanian banks are net receivers on average (−24.1)
- ▣ Lag order: $p = 2$ is held fixed in every window so that the index does not jump when the selected order changes; BIC picks $p = 1$ in 100% of the windows, and with $p = 1$ the index has correlation 0.99 with the one shown

Estimation Error in the Network

- A VAR(1) on 13 series has 169 slope coefficients; a 250-day window gives 250 observations per equation for 13 slopes, and the table is a nonlinear function of all of them
 - ▶ Every number in the table and every arrow of the network is an estimate: it needs an interval
- Residual block bootstrap: resample blocks of 20 days of VAR residual vectors, rebuild the series with the estimated VAR, re-estimate, recompute the table; 999 draws
 - ▶ Resampling whole residual vectors keeps the contemporaneous correlation Σ , which drives the generalized decomposition
- Full sample, $p = 2$: $C = 63.9\%$ with 95% interval [59.9, 65.8]
 - ▶ Window of the peak (250 days to 2020-08-27, $p = 2$): $C = 79.3\%$, interval [65.9, 83.0]; bootstrap mean 75.8%, so small samples also bias C

Net Spillovers with Bootstrap Intervals



- Full-sample VAR(2) on log Parkinson volatility, $H = 10$; 95% intervals from 999 residual block-bootstrap draws (blocks of 20 days)

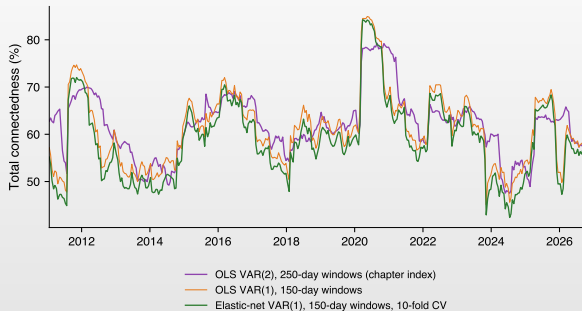
Reading the Net Spillover Intervals

- 8 of 13 net positions have intervals that exclude zero; for the other 5 the label “transmitter” or “receiver” is not identified by the data
- The full sample has about 4 000 days; in the 250-day window of the 2020 peak the interval for C is almost three times wider ($[65.9, 83.0]$), so rolling net labels are mostly noise
- Report the interval next to every net number, and draw only arrows whose pairwise net link is significant
- Frequency connectedness splits the same table into short- and long-horizon parts [Baruník & Křehlík \(2018\)](#); in their study of the global financial crisis, long-term connectedness gains importance after Lehman

Regularised VARs: the Elastic Net

- [Demirer et al. \(2018\)](#), Eq. (9): each VAR equation by the adaptive elastic net, λ chosen equation by equation by 10-fold cross-validation
 - ▶ $\hat{\beta} = \arg \min_{\beta} \sum_t (y_t - \sum_i \beta_i x_{it})^2 + \lambda \sum_i w_i (\frac{1}{2} |\beta_i| + \frac{1}{2} \beta_i^2)$, $w_i = 1/|\hat{\beta}_{i,OLS}|$
 - ▶ Rolling windows of 150 days with the plain elastic net, $w_i = 1$ (Section 5, footnote 15); forecast horizon $H = 10$
- Why: selection and shrinkage in the VAR coefficients, without forcing a sparse **network** (links also come through Σ)
 - ▶ The paper does not state the lag order in the text; we keep the BIC order of our OLS VAR, $p = 2$
- Full sample, adaptive elastic net: $C = 63.3\%$ (OLS 63.9%); 59% of the coefficients are non-zero
 - ▶ With about 4 000 days the penalty hardly matters; it matters in short windows (next slide)

OLS and Elastic-Net Connectedness in Rolling Windows



- Total connectedness every 10 days, $H = 10$: the chapter index (OLS VAR(2), 250 days), OLS VAR(1) and elastic-net VAR(1) on 150-day windows [Demirer et al. \(2018\)](#)

Reading OLS versus Elastic-Net Connectedness

- ▣ Means over 372 windows: 62.0% (chapter index), 61.1% (OLS, 150 days), 58.7% (elastic net, 150 days)
 - ▶ On the same 150-day windows OLS exceeds the elastic net by 2.4 points on average: estimation noise in the coefficients inflates measured connectedness
- ▣ The dynamics agree: correlation 0.82 between the elastic-net index and the chapter index; the crisis peaks are in the same weeks
- ▣ Levels depend on window, estimator and lag order; changes over time are the robust part of the index
- ▣ Long-run partial correlation networks for large panels: [Barigozzi & Brownlees \(2019\)](#)

Granger-Causality Networks and Multiple Testing

- Billio, Getmansky, Lo & Pelizzon (2012) draw an edge $j \rightarrow i$ when j Granger-causes i at 5%; with 13 banks this is $13 \times 12 = 156$ tests
 - ▶ Even with no links at all, about 7.8 edges would appear by chance
- Our tests: bivariate regressions, one lag, heteroskedasticity-robust Wald statistics
 - ▶ Daily returns: 57 edges at 5%, 33 after Holm [Holm \(1979\)](#), 55 with the false discovery rate controlled at 5% [Benjamini & Hochberg \(1995\)](#)
 - ▶ Log volatility, highly persistent: one extra lag that is not tested [Toda & Yamamoto \(1995\)](#), as in the lag-augmented tests of [Shi, Hurn & Phillips \(2020\)](#): 153, 150 and 153 edges
- Holm controls the chance of any false edge under arbitrary dependence; the FDR is the **expected** share of false edges among those drawn, and BH guarantees it for independent tests [Benjamini & Hochberg \(1995\)](#), while tests on overlapping bank pairs are dependent
- Daily returns of banks in different time zones: many “causal” edges only reflect that New York closes after Frankfurt

Recap: Spillover Networks

- ▣ The generalized variance decomposition of a VAR gives a directed, weighted network that does not depend on the ordering
- ▣ Two thirds of bank volatility risk comes from other banks; US banks transmit, HSBC and Romanian banks receive
- ▣ Connectedness jumps in crises and decays slowly: a monitoring tool for supervisors
- ▣ Choices matter: VAR order, horizon, window length, estimator and volatility versus returns (Seminar 18, B5)
- ▣ Every table and every edge has sampling error: bootstrap intervals, shrinkage, and multiplicity control for edge tests

The Idea of the Financial Risk Meter

- For each firm j , a quantile regression at $\tau = 5\%$ of its return on the returns of **all** other firms and on lagged macro variables [Mihoci, Althof, Chen & Härdle \(2020\)](#)
 - ▶ With 100 firms and a 63-day window there are more regressors than days: a penalty is needed
- **LASSO** (Least Absolute Shrinkage and Selection Operator) quantile regression [Belloni & Chernozhukov \(2011\)](#)
 - ▶ $\hat{\beta}_j(\lambda) = \arg \min_{\beta} \frac{1}{n} \sum_t \rho_{\tau} \left(X_{j,t} - \beta \mathbf{o} - \beta^{\top} Z_{-j,t} \right) + \lambda \|\beta\|_1$
 - ▶ λ switches off weak links: a larger λ leaves fewer active coefficients
- The **FRM** (Financial Risk Meter) is the average of the selected λ_j over firms, designed as an indicator of tail interdependence and systemic stress
 - ▶ Built for US financial firms [Mihoci, Althof, Chen & Härdle \(2020\)](#), emerging markets [Ben Amor, Althof & Härdle \(2022\)](#) and China [Wang, Althof & Härdle \(2023\)](#)

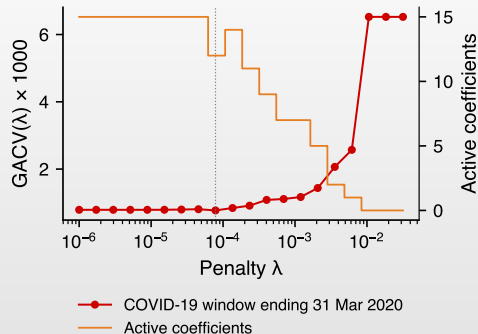
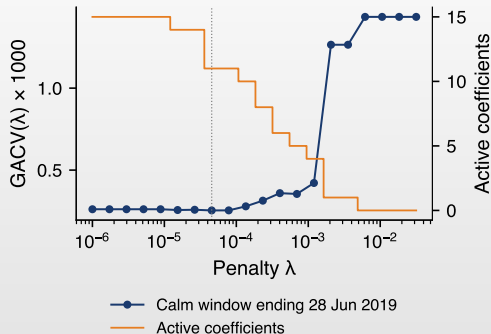
Choosing the Penalty: GACV

- **GACV** (Generalized Approximate Cross-Validation) trades fit against the number of active coefficients

►
$$\text{GACV}(\lambda) = \frac{\sum_t \rho_\tau \left(X_{j,t} - \hat{X}_{j,t}(\lambda) \right)}{n - \text{df}(\lambda)}, \text{ df} = \text{number of nonzero coefficients}$$

- $\hat{\lambda}_j = \arg \min_{\lambda} \text{GACV}(\lambda)$ on a grid; $\text{FRM}_t = \frac{1}{J} \sum_j \hat{\lambda}_{j,t}$
- Our version: 13 banks, macro variables S&P 500, Euro Stoxx 50 and VIX at $t - 1$, 63-day windows every 10 days
 - 15 regressors for 63 days: far fewer than in the published FRM
- Extensions: FRM based on expectiles [Ren, Lu, Li & Härdle \(2022\)](#), for cryptocurrencies [Ren, Althof & Härdle \(2022\)](#) and for the EU energy sector [Toma, Dima, Grecu, Pele & Lessmann \(2026\)](#)

GACV in a Calm and in a Crisis Window

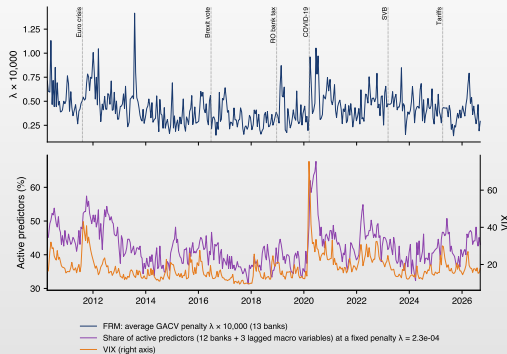


- JPMorgan regressed on the other 12 banks and three lagged macro variables, 63 days; dotted line: selected λ; orange: active coefficients

Reading the GACV Curves

- Calm window (to 28 June 2019): $\hat{\lambda} = 4.5 \times 10^{-5}$, 11 of 15 coefficients active
 - ▶ COVID-19 window (to 31 March 2020): $\hat{\lambda} = 7.8 \times 10^{-5}$, 12 active
- The curve is flat for small λ and rises steeply once key links are switched off
- In the crisis the whole curve is about three times higher: the check loss is in return units, and returns are larger
- With few regressors, the minimum lies where almost no penalty is applied: the selection is noisy

The Financial Risk Meter of Thirteen Banks



- Top: average GACV penalty over 13 banks; bottom: share of active predictors (12 banks and 3 lagged macro variables) when λ is held fixed, with the VIX; 390 windows of 63 days

Reading Our Financial Risk Meter

- The average GACV penalty is noisy; its correlation with the VIX is only 0.35
 - ▶ With 15 regressors and 63 days the fit hardly needs a penalty; the published FRM uses about 100 firms, so that the penalty must do real work
- At a fixed penalty $\lambda = 2.3 \times 10^{-4}$, the share of active predictors (bank links and macro variables) follows the VIX more closely: correlation 0.62
 - ▶ From 31% (2017-11-21) to 68% (2020-06-17); average 42%
- Lesson: this small-panel application does not show that either the selected penalties or a fixed-penalty count of active predictors is a reliable systemic-risk indicator; the fixed-penalty count is only an exploratory sensitivity check

Recap: the Financial Risk Meter

- Each firm's 5% quantile is regressed on all other firms with an L1 penalty; the penalty λ chosen by GACV summarises tail interdependence
- The FRM is designed to signal systemic stress; it relies on large cross-sections of firms
- On 13 banks the penalty is noisy; at a fixed penalty the share of active predictors co-moves with the VIX, without evidence that it predicts stress
- Advanced material: Quantinar course on the FRM (end of chapter)

Climate Risk as a Systemic Risk

- ▣ **Physical risk:** losses from floods, droughts, heat and storms that damage collateral and borrowers
 - ▶ **Transition risk:** losses from the move to a low-carbon economy (carbon prices, regulation, technology)
- ▣ “Green swan” risks: potentially extremely disruptive events that could be behind the next systemic financial crisis; backward-looking risk models cannot anticipate them [Bolton et al. \(2020\)](#)
- ▣ A network climate stress test maps investors’ exposures to climate-policy-relevant sectors and adds second-round effects through the financial network [Battiston et al. \(2017\)](#)
- ▣ Hence scenario analysis instead of historical estimation: the past sample contains no energy transition

The ECB Climate Risk Stress Test of 2022



Seat of the European Central Bank, Frankfurt, 2015

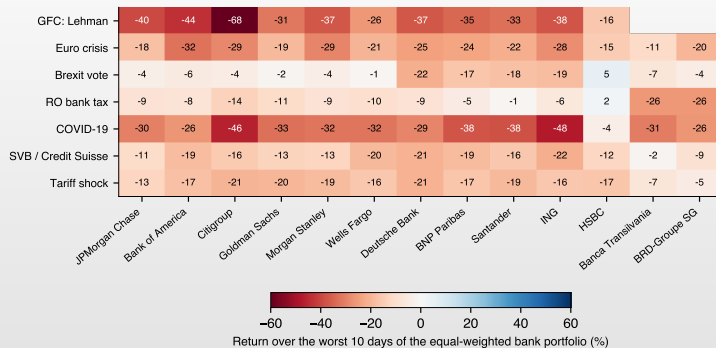
Photo: DXR (2015); CC BY-SA 4.0; Wikimedia Commons

- 104 significant banks took part; a bottom-up projection of losses covered 41 of them [ECB Banking Supervision \(2022\)](#)
- About 60% of banks had no climate risk stress-testing framework; only 20% considered climate risk when granting loans
- Almost two thirds of banks' income from non-financial corporate customers came from greenhouse gas-intensive industries
- Credit and market losses of around EUR 70 billion for the 41 banks, under a short-term disorderly transition and two physical risk scenarios
 - ▶ The ECB states that this significantly understates the risk: data gaps, rudimentary models, no second-round effects

Stress Tests and Historical Scenarios

- A stress test asks how much we would lose in a **severe but plausible** scenario: **historical** (replay a past crisis), **hypothetical** (shock chosen factors), **reverse** (find the most plausible scenario behind a fatal loss)
 - ▶ Supervisory tests of large banks: [Federal Reserve stress tests](#) in the US, EU-wide tests of the [European Banking Authority](#); global systemically important banks hold extra capital [Basel Committee \(2018\)](#)
- Seven episodes, each a window of a few weeks around a known shock
 - ▶ Lehman (autumn 2008), the euro crisis (summer 2011), the Brexit vote (June 2016), the Romanian bank tax (December 2018), COVID-19 (March 2020), SVB and Credit Suisse (March 2023), the tariff shock (April 2025)
- Inside each window we find the **worst 10 trading days** of the equal-weighted bank portfolio and report every bank's return over those days
- The 2008 episode uses the 11 international banks only: the Romanian series start in 2010
- Ten days match the horizon of market-risk capital (Chapter 7)

Seven Historical Stress Scenarios



Return (%) of each bank over the worst 10 days of the equal-weighted bank portfolio in each episode; blank: no data

Reading the Historical Scenarios

- Portfolio loss over the worst 10 days: Lehman 37.6% (2008-11-10 to 2008-11-21), COVID-19 31.9%, euro crisis 22.3%
 - ▶ Tariffs 16.0%, SVB 14.8%, Romanian bank tax 10.2%, Brexit 7.9%
- Worst single cells: Citigroup -68% in 2008, ING -48% in March 2020
- Shocks are local as well as global: Brexit hit mainly European banks; the Romanian bank tax hit Romanian banks hardest (-26% and -26%), during a December 2018 sell-off that also cost US banks -8% to -14%
- A stress test built only on global crises would miss the domestic political shock

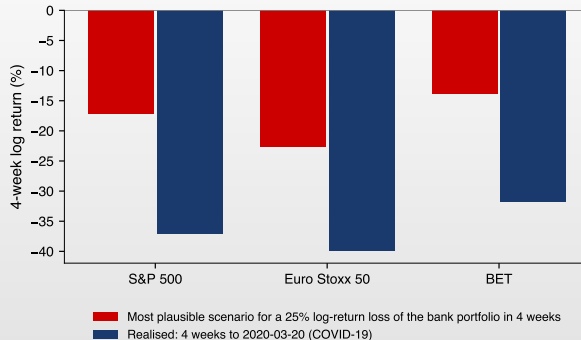
Reverse Stress Testing

- Factor model: weekly bank log returns (%) $X_i = \beta_i^\top f + e_i$, factors $f = (\text{S\&P 500, Euro Stoxx 50, BET})$, over h weeks $f \sim N(0, h\Sigma)$ (no serial covariance of weekly factor returns)
 - ▶ Loss of the equal-weighted portfolio in log-return points: $L = -b^\top f$, with b the average of the β_i ; a factor-only approximation that sets the residual $\sum_i w_i e_i$ to zero
- The most plausible scenario with loss ℓ^* minimises the Mahalanobis distance $f^\top (h\Sigma)^{-1} f$ subject to $-b^\top f = \ell^*$
 - ▶ $f^* = -\ell^* \frac{h\Sigma b}{b^\top h\Sigma b}, \quad d^* = \frac{\ell^*}{\sqrt{b^\top h\Sigma b}}$
- d^* counts portfolio standard deviations: the larger it is, the less plausible the loss under the model

Worked Example: Reverse Stress Test with Two Factors

- Betas $b = (1.2, 0.8)$; four-week covariance of the factors $h\Sigma = \begin{pmatrix} 16 & 6 \\ 6 & 9 \end{pmatrix}$ (%²); target loss $\ell^* = 20\%$
 - ▶ $h\Sigma b = (1.2 \cdot 16 + 0.8 \cdot 6, ; 1.2 \cdot 6 + 0.8 \cdot 9) = (24, 14.4)$
 - ▶ $b^\top h\Sigma b = 1.2 \cdot 24 + 0.8 \cdot 14.4 = 40.32$, standard deviation 6.35%
- Most plausible scenario and its distance
 - ▶ $f^* = -20 \cdot (24, 14.4)/40.32 = (-11.9\%, -7.1\%)$, $d^* = 20/6.35 = 3.15$
 - ▶ Normal probability of a loss at least this large: $\Phi(-3.15) = 0.08\%$ per four weeks
- The scenario loads more on the factor with the larger beta and variance, and both factors fall together because they are correlated

The Most Plausible Scenario for a 25% Log-Return Loss



- Weekly data 2010–2026 (871 weeks), four-week horizon; red: f^* for a log-return loss $\ell^* = 25\%$; blue: factor moves in the worst four weeks of 2020

Reading the Reverse Stress Test

- Portfolio betas: $b = (0.45, 0.65, 0.18)$; four-week standard deviation 5.79%
 - ▶ A log-return loss of 25% (an arithmetic loss of 22.1%) needs log returns of S&P 500 -17.2% , Euro Stoxx 50 -22.6% , BET -13.9% together
- Distance $d^* = 4.31$: under the Normal model its probability is 8.0×10^{-6} per four weeks, practically impossible
- Yet in the four weeks to 2020-03-20 the portfolio lost 52.0% in log terms, with factor falls larger than f^* (the model implies 48.5%)
- The shape of the scenario is informative; its probability is not: heavy tails and rising correlations make Normal probabilities far too small

Limits of Stress Tests

- ▣ Scenarios are chosen by people: the crisis that comes is rarely the one that was tested
- ▣ Second-round effects (fire sales, funding runs, counterparty losses) are hard to model and usually left out, as the ECB noted for its climate test
- ▣ Market-based measures (MES, CoVaR, connectedness) react quickly but only after prices move; balance-sheet stress tests look forward but slowly
- ▣ Early-warning research combines both, e.g. entropy-based measures of market risk [Pele, Lazar & Dufour \(2017\)](#)

Recap: Stress Testing

- ▣ Historical, hypothetical and reverse stress tests answer different questions
- ▣ Across seven episodes, the worst 10-day loss of a bank portfolio ranged from 7.9% to 37.6%
- ▣ Reverse stress tests give the most plausible direction of a disaster, not its probability
- ▣ Climate scenarios extend stress testing to risks that have no history

Case Study: Question and Motivation

- [Pele, Conda, Bag, Mazurencu-Marinescu-Pele & Strat \(2023\)](#) build FRM@RO, the Financial Risk Meter of the BVB: can the average GACV penalty of Section 5 gauge systemic risk in a small emerging market?
 - ▶ Which macro factors drive the joint tail moves of Romanian stocks?
 - ▶ Does the index lead market volatility and the business cycle?
- Why the BVB: re-established in 1995, upgraded by FTSE Russell from Frontier to Secondary Emerging market in 2020 (Section 1)
 - ▶ In 2008 the Romanian market was the most affected in Central and Eastern Europe (Section 3.2)
- Link to this chapter: the quantile LASSO coefficients form a daily **tail network** (adjacency matrix, Eq. 7), the tail counterpart of the spillover networks of Section 4
 - ▶ A higher λ_j signals stronger joint tail risk of firm j with the rest of the market (Section 2.1)
- Same framework as for US and European firms [Mihoci, Althof, Chen & Härdle \(2020\)](#), emerging markets [Ben Amor, Althof & Härdle \(2022\)](#) and China [Wang, Althof & Härdle \(2023\)](#)

Case Study: Data and Method as Published

- Data (Section 3.1, Tables A1–A2): daily closing prices and market capitalisation of 61 BVB companies, 3 January 2005 – 13 December 2022 (4 498 trading days); log returns, first differences for interest rates
 - ▶ 14 macro variables: BET, BET-FI, BET-NG, ROTX; S&P 500, Euro Stoxx 50, DAX, CAC 40; VIX; S&P US Treasury Bill Index; ROBOR 3M, EURIBOR 3M; the leu against the euro and the US dollar
- Model (Section 2.1, Eqs. 1–6; Section 3.2): each day the $J = 20$ largest companies by market capitalisation; for each j , a quantile LASSO regression at $\tau = 0.05$ on the 14 macro variables and the other 19 companies, rolling window $n = 63$ days
 - ▶ λ_j chosen by GACV; $\text{FRM@RO}_s = \frac{1}{J} \sum_{j=1}^J \lambda_j^s$; 33 regressors for 63 days, so the penalty does real work
- Influence of a macro variable (Eq. 8): the share of the J companies whose regression keeps it with a nonzero coefficient
- Backtests (Sections 2.4 and 3.6): Granger causality, maximum lag 5, between FRM@RO and the GARCH(1,1) volatility of BET
 - ▶ Logit of a technical-recession indicator (two consecutive quarterly falls of seasonally adjusted real gross domestic product) on the quarterly average FRM@RO lagged $k = 1, 2, 3$ quarters; Nagelkerke pseudo- R^2

Case Study: Key Results

- Mean influence, share of companies (Table 2): VIX 80.87%, ROBOR 3M 66.82%, BET-FI 22.09%, BET-NG 21.04%, DAX 14.92%
 - ▶ Global fear dominates the tails; domestic funding costs and the investment-fund and energy sectors come next
- Granger causality with BET volatility (Table 3, $\tau = 0.05$): volatility to FRM@RO $F = 20.68$, FRM@RO to volatility $F = 12.35$, both $p < 0.001$; at $\tau = 0.01$: $F = 14.84$ and $F = 7.67$, both $p < 0.001$
 - ▶ Feedback in both directions: FRM@RO follows market volatility and also helps predict it
- Logit on technical recession (Table 5), slope (standard error) and pseudo- R^2 : lag 1: 168.537 (59.324), 0.482; lag 2: 118.578 (44.141), 0.318; lag 3: 97.115 (39.115), 0.234
 - ▶ The signal is strongest one quarter ahead and fades with the horizon
- Episodes (Figures 2 and 7): peaks in 2008, in June 2010 (public-sector wages cut by 25%), in March 2020 and in 2022 (war in Ukraine)
 - ▶ High predicted recession probabilities in 2009Q1–Q2 and 2010Q3, and a warning before the 11.17% fall of real output in 2020Q2

Case Study: Replicate, Limits, Extend

- Replicate: data (prices, market capitalisations, macro variables), the R code `FRM_Quantlet_R0.R` and the daily FRM@RO series with risk levels at danpele.github.io/frm_ro
 - ▶ Backtests step by step: GARCH(1,1) on BET log returns (`data/market/BET.csv`), Granger F tests with lags 1 to 5 against the published FRM@RO series, logit on quarterly averages
- Course data to 2026: 11 BVB stocks (banks, energy and utilities, Fondul Proprietatea, Digi) and BET, BET-FI, BET-NG, VIX, S&P 500, Euro Stoxx 50, DAX
 - ▶ Fewer than $J = 20$ companies: a data limitation to state, not a reason to change the method
- Limits of the method: λ_j measures joint tail moves, not causal contagion (a common shock moves every node); Granger causality is predictive precedence
 - ▶ Further research proposed by the authors (Section 5): an integrated early-warning system with unemployment, exchange rates and quarterly output
- Open extension: out of sample, 2023–2026, does FRM@RO still lead BET volatility, and which macro variables drive the tails after the 2025 tariff shock?

An Open Question

Does network connectedness measure systemic risk, or only volatility?

- Connectedness indices are monitored as early-warning tools, but an index built from a VAR on volatilities may co-move with volatility without adding predictive information
 - ▶ Live research: global bank networks from regularised VARs [Demirer et al. \(2018\)](#); connectedness in the tails differs from connectedness in the mean [Ando, Greenwood-Nimmo & Shin \(2022\)](#)
- Formal version, for future bank stress y_{t+h} , total connectedness C_t and current volatility σ_t :
$$y_{t+h} = a + \beta_C C_t + \gamma \ln \sigma_t + e_{t+h}, \quad H_0: \beta_C = 0, \quad H_1: \beta_C > 0$$
 - ▶ Falsified if C_t loses its predictive power once current volatility is controlled
- Identification: a network edge is a share of forecast error variance, not a causal exposure
 - ▶ A common shock raises volatility and connectedness together, with no bank infecting another

The Discovery Loop, Step by Step (1/2)

- **Literature:** map what is known, then verify every reference
 - ▶ “Which papers test whether connectedness indices predict bank distress out of sample, beyond volatility? Give DOI, sample, horizon and controls.”
 - ▶ Resolve each DOI on Crossref and check that the paper really controls for volatility; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses:** competing readings of a rise in connectedness, each with its own test
 - ▶ “Give competing interpretations of a rise in total connectedness (common shock, contagion, artefact of rolling windows) and a test that discriminates between them.”
 - ▶ E.g. a common shock fades with volatility; contagion has a direction: shocks of net transmitters precede those of receivers
- **Data and code:** on the course data, replicating a known number first
 - ▶ “Fit a VAR(2) to the log Parkinson volatilities of the 13 banks in data/market on 250-day windows and compute the generalized total connectedness with $H = 10$.”
 - ▶ Replicate first: mean 62.0% and maximum 79.3% (2020-08-27), as in this chapter

The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed before the predictive regressions are run
 - ▶ “Pre-register: horizons 21/63/126 days, volatility and VIX controls, windows 150/250/500 days, HAC lags at least the overlap, out-of-sample evaluation from 2018.”
 - ▶ Every cell of the grid is reported; the family is corrected with Holm or controlled with the FDR
- **Critique**: ask the assistant to attack the result
 - ▶ “Act as a referee: is the predictive power of connectedness just the persistence of volatility? Which regression would show it?”
 - ▶ Expect: add current volatility as a control, then test the increment of R^2 and of the out-of-sample loss
- **Report**: standardised effects with HAC t -statistics, every cell of the grid, what failed
 - ▶ The prompts that shaped the design go into AI_USE.md

What the Researcher Must Check

- **Non-stationary inputs:** a VAR on price levels, or on trending series, produces spurious connectedness
 - ▶ Test: unit-root tests on every input; all eigenvalues of the companion matrix of the fitted VAR strictly inside the unit circle in every window
- **Ordering:** Cholesky decompositions change with the order of the banks
 - ▶ Test: permute the columns; the generalized table must not change
- **Look-ahead:** a window labelled by its first date, or a lag order chosen on the full sample
 - ▶ Test: the index at t must not move when data after t are deleted
- **Overlap:** rolling indices and forward outcomes overlap, so OLS t -statistics are inflated
 - ▶ Test: Newey–West lags at least equal to the overlap, or a block bootstrap
- **Dimension:** 13 banks and $p = 2$ give 27 coefficients per equation from 250 days; more banks need shrinkage [Demirer et al. \(2018\)](#)
 - ▶ Test: compare with a LASSO VAR; unstable edges signal estimation noise, not structure

Mini-Case: Is Connectedness More Than Volatility?

- A typical assistant proposal: “high total connectedness warns of future bank stress”; we test it on 731 weekly dates, 2011–2026
 - ▶ y_{t+h} : log mean Parkinson volatility of the 13 banks over the next $h = 63$ days; control: the same over the last 20 days; standardised variables, Newey–West with 13 lags
- Connectedness alone: $\hat{\beta}_C = 0.40$ ($t = 4.63$), $R^2 = 0.16$: the proposal looks right
- With current volatility: $\hat{\beta}_C = 0.08$ ($t = 0.75$), $\hat{\gamma} = 0.53$ ($t = 4.13$); $R^2 = 0.343$ vs 0.339 with volatility alone
 - ▶ Correlation of C_t with current volatility: 0.59; for the next 63-day maximum drawdown of the bank portfolio, $\hat{\beta}_C = -0.10$ ($t = -1.32$) even alone
- Interpretation: once current volatility is controlled, the increment of connectedness is not significant; this does not prove it is zero, but in our sample the persistence of volatility accounts for most of the predictive content; network information may still matter in the tails [Ando, Greenwood-Nimmo & Shin \(2022\)](#) or for the direction of transmission

Your Turn: a Project Seed

- **Question:** does **tail** connectedness predict bank distress beyond volatility, out of sample?
- **Replicate first:** our rolling index (mean 62.0%, spring 2020 78.4%), then one headline connectedness figure of [Demirer et al. \(2018\)](#) on their sample
 - ▶ Extend: quantile connectedness at $\tau = 5\%$ [Ando, Greenwood-Nimmo & Shin \(2022\)](#), out-of-sample evaluation from 2018, MES and SRISK as alternative targets
- **Deliverables:** a pre-registered grid committed before the regressions, a notebook that regenerates every number, HAC inference and out-of-sample losses for every cell
- **AI log:** prompts that shaped the design in AI_USE.md; each wrong claim (a VAR on price levels, OLS t -statistics on overlapping windows) and the test that caught it in AI_ERRORS.md ([templates](#))

Which Bank Is Most Systemic?

Bank	VaR 1%	MES 5%	$\Delta\text{CoVaR 1\%}$	Net spillover	L^*
JPMorgan	4.83	3.22	4.62	9.8	7.0
Bank of America	5.99	3.85	4.60	10.9	6.2
Citigroup	5.87	4.03	3.81	19.0	6.0
Goldman Sachs	4.81	3.34	4.12	2.4	6.8
Morgan Stanley	5.83	4.04	4.00	9.9	6.0
Wells Fargo	5.78	3.32	4.89	3.6	6.8
Deutsche Bank	7.19	4.54	4.24	-12.6	6.1
BNP Paribas	6.18	4.53	4.84	6.0	6.1
Santander	5.83	4.23	4.42	-5.4	6.4
ING	7.16	4.81	4.42	5.2	5.8
HSBC	4.66	2.54	4.07	-23.6	8.3
Banca Transilvania	5.06	3.14	—	-12.2	6.5
BRD-Groupe SG	4.42	2.69	—	-13.0	7.1

□ Daily losses in %; net spillover of volatility (to minus from); L^* : break-even leverage with $k = 8\%$

Reading the Comparison

- ▣ Rank correlations across the eleven international banks
 - ▶ MES and ΔCoVaR : -0.01 ; MES and net spillover: 0.04 ; ΔCoVaR and net spillover: -0.01
 - ▶ VaR and MES: 0.88
- ▣ The measures answer different questions: exposure to the market, contribution to the system, transmission of volatility
- ▣ No single ranking: supervisors combine several indicators with size and balance-sheet data, as in the [Basel Committee \(2018\)](#) methodology
- ▣ Romanian banks: moderate MES, net receivers of volatility, break-even leverage close to that of US banks

Key Takeaways

- ▣ Systemic risk lives in the joint tail: exposure (MES, SRISK), contribution (ΔCoVaR) and connectedness (spillover and tail networks)
- ▣ European banks have the largest MES in our sample; US banks are the main net transmitters of volatility
- ▣ VaR and ΔCoVaR rank banks differently; rankings are uncertain: test differences, report intervals, state αn
- ▣ Market-based measures jump in crises; stress tests and balance sheets are needed to see risk building up
- ▣ Romanian banks are mostly driven by domestic shocks and receive more than they transmit

Common Pitfalls

- ▣ Reading ΔCoVaR as causal: a common shock creates the same pattern
- ▣ Reading overlapping intervals as “equal” and separate ones as “different”: test the difference on joint draws, and correct for the number of pairs
- ▣ Including the bank itself in the system portfolio: its own return adds a mechanical self-contribution to the link
- ▣ Orthogonal variance decompositions whose results depend on the order of the banks
- ▣ Reporting SRISK without balance-sheet data, or from the $1 - e^{-18\text{MES}}$ shortcut as if it were the published measure
- ▣ Drawing every Granger edge significant at 5% out of 156 tests, or every net label without its interval

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Financial Risk Meter for Emerging Markets
- ▣  Dynamic Crypto Networks
- ▣  Network Centrality

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_18

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